

In-season estimates of ad-intact fish returning to the Krassel PIT array

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Summary

- Current PIT-tag expansions at KRS (as of July 17th):
 - 655 natural-origin fish estimated from 118 observed tags
 - 560 integrated fish estimated from 80 tags with a ratio of approximately 7 fish/tag
- KRS was non-operational: 7/10, 7/11, 7/14, 7/15, and 7/16
 - Integrated and natural-origin PIT-tag passage during this period is estimated as ≤ 10 for each group.
 - additional 70 integrated fish
 - additional 56 natural-origin
- Ad-intact fish currently estimated above KRS: 1,341
- The average passage date of $\geq 75\%$ of the total return across KRS is July 15th (Figure 1).
- End of season ad-intact total estimate: 1,788
 - NOR: $(655 + 56)/.75 = 948$
 - Integrated: $(560 + 70)/.75 = 840$

Assumptions

- 100% detection efficiency at KRS
- Static interpolation rate of 2 tags when observations are typically decreasing.
- Integrated fish observed at KRS is approximately 100 fish higher than the estimate at LGR.
- Integrated tag ratio of 7 is slighter higher than actual rate for each release group.

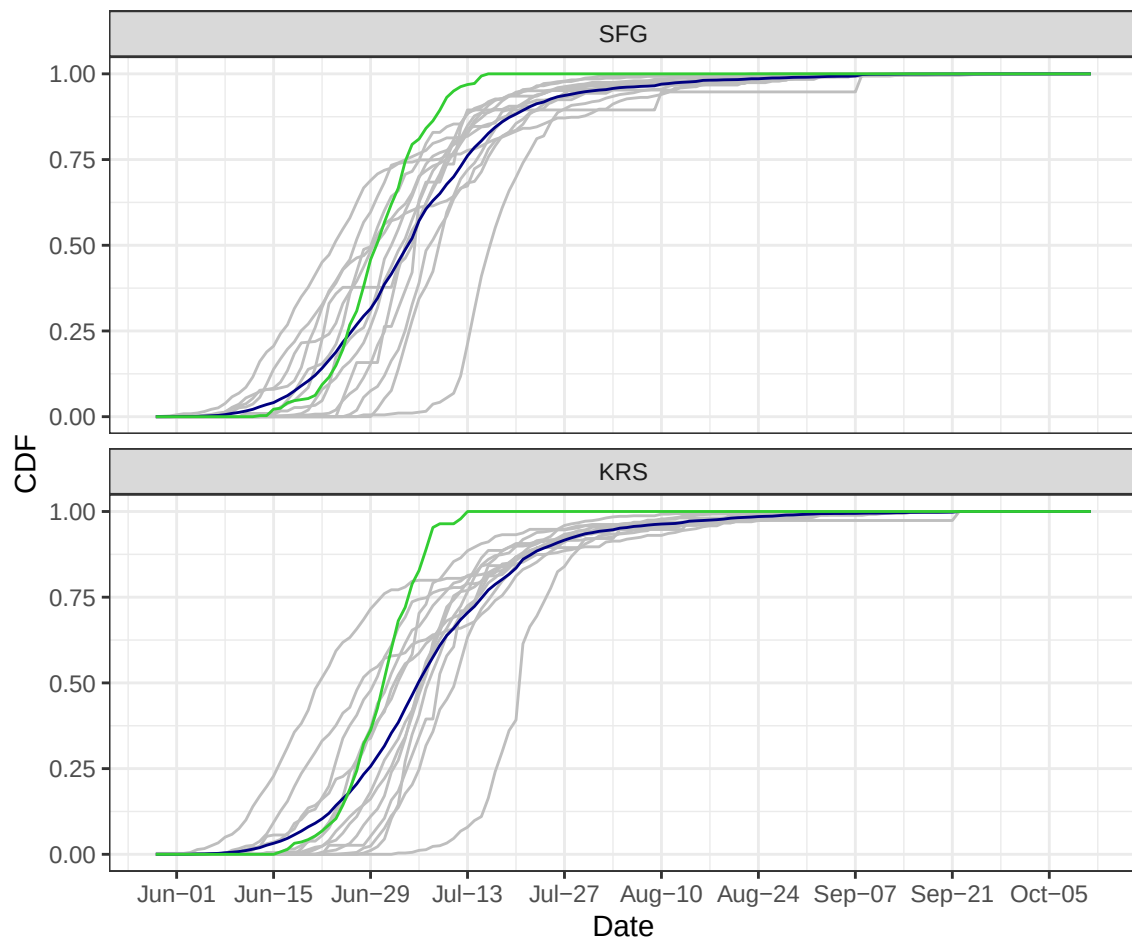


Figure 1: Run-timing of SFSR hatchery (segregated and integrated) and natural origin returns across PIT-arrays. The grey lines indicate individual spawn year returns for 2010-2022, the dark blue line is the average cumulative proportion of returns across the same years, and the green line represents the current 2023 return.